

Week 6

Database Design Principles

Data Normalization in Database Design

Normalization

- Question:
- If you have a table with student records, and you store the student's department name with every student, what problems might happen if the department name changes?"

Normalization

- Helps deal with data redundancy and insert, update and delete anomalies. Anomalies are problems that can occur in poorly planned, unnormalized databases where all the data is stored in one table (a flat-file database).
- **Normalization:** Organize data in a database to reduce redundancy and improve data integrity.
- This is done by applying normal forms (1NF, 2NF, 3NF, BCNF, etc)

Problems Normalization Solves

- **Data Redundancy:** Storing same data multiple times.
- **Update Anomaly:** Updating value in one place but not in another resulting in inconsistencies.
- **Insertion Anomaly:** Unable to insert because of missing other info.
- **Deletion Anomaly:** the unintended loss of data due to deletion of other data.

Example Table (Unnormalized)

StudentID	StudentName	DeptName	DeptHead
1	Alice	CS	Dr. Smith
2	Bob	CS	Dr. Smith
3	Carol	Math	Dr. Lee

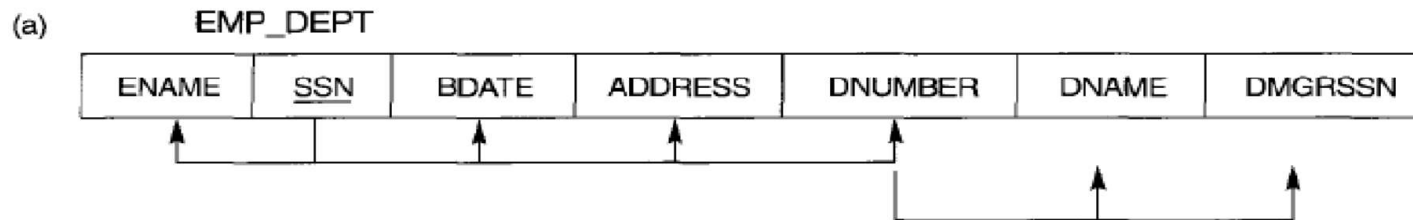
- Redundancy:** DeptName + DeptHead repeated.
- Update Anomaly:** If “CS Head” changes → must update many rows.
- Insertion Anomaly:** Can’t insert a new department until a student joins.
- Deletion Anomaly:** If all CS students leave → lose info about CS dept.

Functional Dependencies

- **Functional Dependency(FD):** is a relationship that exists when one attribute uniquely determines another attribute.
- An attribute Y is said to have a functional dependency on a set of attributes X (written $X \rightarrow Y$) if and only if each X value is associated with precisely one Y value.
- It's represented by an arrow **sign ($\rightarrow$)** that is, $X \rightarrow Y$, where X functionally determines Y. The left-hand side attributes determine the values of attributes on the right-hand side.
- Customarily we call **X the determinant** set and **Y the dependent** attribute.

Functional Dependencies

Example:



$F = \{SSN \rightarrow \{ENAME, BDATE, ADDRESS, DNUMBER\},$
 $DNUMBER \rightarrow \{DNAME, DMGRSSN\}\}$

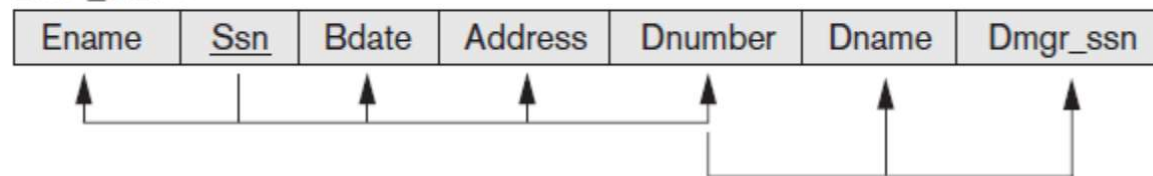
Two relational schemas suffering from anomalies

Figure 15.3

Two relation schemas suffering from update anomalies. (a) EMP_DEPT and (b) EMP_PROJ.

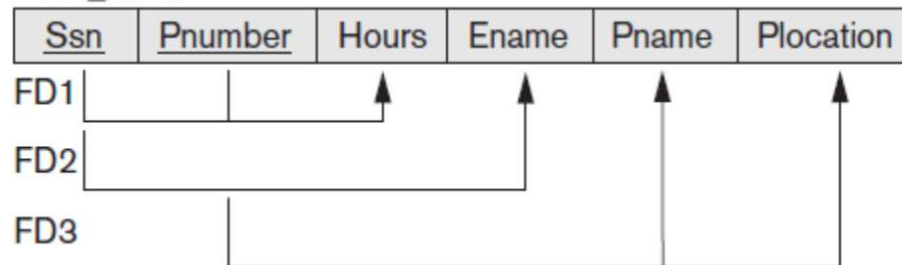
(a)

EMP_DEPT



(b)

EMP_PROJ



					Redundancy	
EMP_DEPT						
Ename	Ssn	Bdate	Address	Dnumber	Dname	Dmgr_ssn
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 FireOak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555

			Redundancy		Redundancy	
EMP_PROJ						
Ssn	Pnumber	Hours	Ename	Pname	Plocation	
123456789	1	32.5	Smith, John B.	ProductX	Bellaire	
123456789	2	7.5	Smith, John B.	ProductY	Sugarland	
666884444	3	40.0	Narayan, Ramesh K.	ProductZ	Houston	
453453453	1	20.0	English, Joyce A.	ProductX	Bellaire	
453453453	2	20.0	English, Joyce A.	ProductY	Sugarland	
333445555	2	10.0	Wong, Franklin T.	ProductY	Sugarland	
333445555	3	10.0	Wong, Franklin T.	ProductZ	Houston	
333445555	10	10.0	Wong, Franklin T.	Computerization	Stafford	
333445555	20	10.0	Wong, Franklin T.	Reorganization	Houston	
999887777	30	30.0	Zelaya, Alicia J.	Newbenefits	Stafford	
999887777	10	10.0	Zelaya, Alicia J.	Computerization	Stafford	
987987987	10	35.0	Jabbar, Ahmad V.	Computerization	Stafford	
987987987	30	5.0	Jabbar, Ahmad V.	Newbenefits	Stafford	
987654321	30	20.0	Wallace, Jennifer S.	Newbenefits	Stafford	
987654321	20	15.0	Wallace, Jennifer S.	Reorganization	Houston	
888665555	20	Null	Borg, James E.	Reorganization	Houston	

In EMP_DEPT, the attribute values pertaining to a particular department (DNUMBER, DNAME, DMGRSSN) are repeated for every employee who works for that department.

Update anomaly

- In **EMP_DEPT**, if we change the value of one of the attributes of a particular department, For example the manager of department 5
 - we must update the tuples of all employees who work in that department; otherwise, the database will become inconsistent.
- If we **fail to update** some tuples, the same department will be shown to have two different values for manager in different employee tuples, which would be wrong.

Insert Anomaly

- can be differentiated into two types, based on the **EMP_DEPT** relation:
- To insert a new employee tuple into **EMP_DEPT**, we must include either the attribute values for the department that the employee works for
- or **nulls** (if the employee does not work for a department as yet).

EMP_DEPT					Redundancy	
Ename	Ssn	Bdate	Address	Dnumber	Dname	Dmgr_ssn
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 FireOak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555

Insert Anomaly

- It is difficult to insert a new department that has no employees as yet in the **EMP_DEPT** relation.
- The only way to do this is to place **null values** in the attributes for employee.
- This causes a problem because **SSN is the primary key** of EMP_DEPT, and each tuple is supposed to represent an employee entity-not a department entity.

EMP_DEPT						Redundancy
Ename	Ssn	Bdate	Address	Dnumber	Dname	Dmgr_ssn
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
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Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555

Delete Anomaly

- If we delete from **EMP_DEPT** an employee tuple that happens to represent the last employee working for a particular department, the information concerning that department is lost from the database.

							Redundancy
EMP_DEPT							
Ename	Ssn	Bdate	Address	Dnumber	Dname	Dmgr_ssn	
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555	
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555	
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321	
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321	
Narayan, Ramesh K.	666884444	1962-09-15	975 FireOak, Humble, TX	5	Research	333445555	
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555	
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321	
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555	

						Redundancy	Redundancy
EMP_PROJ							
Ssn	Pnumber	Hours	Ename	Pname	Plocation		
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123456789	2	7.5	Smith, John B.	ProductY	Sugarland		
666884444	3	40.0	Narayan, Ramesh K.	ProductZ	Houston		
453453453	1	20.0	English, Joyce A.	ProductX	Bellaire		
453453453	2	20.0	English, Joyce A.	ProductY	Sugarland		
333445555	2	10.0	Wong, Franklin T.	ProductY	Sugarland		
333445555	3	10.0	Wong, Franklin T.	ProductZ	Houston		
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987654321	30	20.0	Wallace, Jennifer S.	Newbenefits	Stafford		
987654321	20	15.0	Wallace, Jennifer S.	Reorganization	Houston		
888665555	20	Null	Borg, James E.	Reorganization	Houston		

Better design

- These problems don't occur in this design because DEPARTMENT tuples are stored separately.

EMPLOYEE

Ename	<u>Ssn</u>	Bdate	Address	Dnumber
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4
Narayan, Ramesh K.	666884444	1962-09-15	975 Fire Oak, Humble, TX	5
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1

DEPARTMENT

Dname	<u>Dnumber</u>	Dmgr_ssn
Research	5	333445555
Administration	4	987654321
Headquarters	1	888665555

Superkeys, Candidate keys, Primary key

- **Superkeys:** Any set of attributes that can uniquely identify a row.
- **Candidate keys:** a minimal superkey. If you can remove an attribute from a superkey and it's still a superkey → then it's not minimal.
- Example: **Relation:** R(StudentID, Email, Name, Address,...)
- **Possible superkeys:** {StudentID}, {Email}, {StudentID, Email}, {StudentID, Name}, {Email, Name}, {StudentID, Email, Name}
- **Test minimality**
 - {StudentID} unique → candidate key
 - {Email} unique → candidate key
 - {StudentID, Email} not minimal (remove one, still unique)
 - {StudentID, Name} not minimal (StudentID alone works)
 - {Email, Name} not minimal (Email alone works)
 - {StudentID, Email, Name} not minimal
- **Candidate Keys** = {StudentID}, {Email}.

Normalization

- Takes a relation schema through a series of tests
- Normal forms (NF): provide criteria for determining a table's degree of vulnerability to logical inconsistencies and anomalies.
- Pays particular attention to normalization only up to 3NF, BCNF, or at most 4NF
- Do not need to normalize to the highest possible normal form

First Normal Form (1NF)

- **Condition 1:** 1 NF does not allow:
 - duplicate rows.
 - Composite attributes
 - Multivalued attributes
- **Condition 2:** Every row-and-column intersection contains exactly one value (and nothing else).

First Normal Form (1NF)

- Example of tables that would NOT meet this definition of 1NF are:
- A table that **lacks a unique key**. Such a table would be able to accommodate duplicate rows, in violation of condition 1.
- A table with at least one **nullable attribute**. A nullable attribute would be in violation of condition 2, which requires every field to contain exactly one value from its column's domain.

First Normal Form (1NF)

- Example: Suppose a designer wishes to record the names and telephone numbers of customers. He defines a customer table which looks like this:

Customer			
Customer ID	First Name	Surname	Telephone Number
123	Robert	Ingram	555-861-2025
456	Jane	Wright	555-403-1659 555-776-4100
789	Maria	Fernandez	555-808-9633

First Normal Form (1NF)

- Example:
- The designer might attempt to get around this restriction by defining multiple Telephone Number columns:

Customer					
Customer ID	First Name	Surname	Tel. No. 1	Tel. No. 2	Tel. No. 3
123	Robert	Ingram	555-861-2025		
456	Jane	Wright	555-403-1659	555-776-4100	555-403-1659
789	Maria	Fernandez	555-808-9633		

- This representation allows use of nullable columns, and therefore does not conform to definition of 1NF and causes logical problems.

First Normal Form (1NF)

- A design that is in 1NF makes use of two tables:
- a Customer Name table and
- a Customer Telephone Number table.

Customer Name		
Customer ID	First Name	Surname
123	Robert	Ingram
456	Jane	Wright
789	Maria	Fernandez

Customer Telephone Number	
Customer ID	Telephone Number
123	555-861-2025
456	555-403-1659
456	555-776-4100
789	555-808-9633

Second Normal Form (2NF)

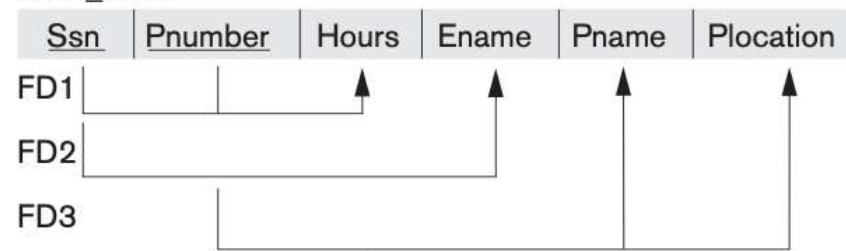
- A table that is in first normal form (1NF) must meet additional criteria if it is to qualify for second normal form.
- **Rule:** Must be in 1NF + **no partial dependency** (non-key attributes must depend on the whole primary key, not part of it).

Second Normal Form (2NF)

Example:

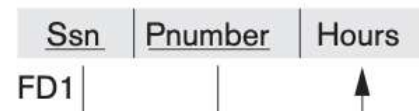
(a)

EMP_PROJ

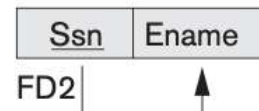


2NF Normalization

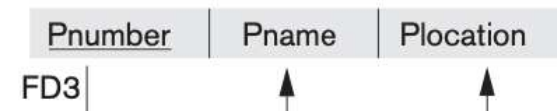
EP1



EP2



EP3



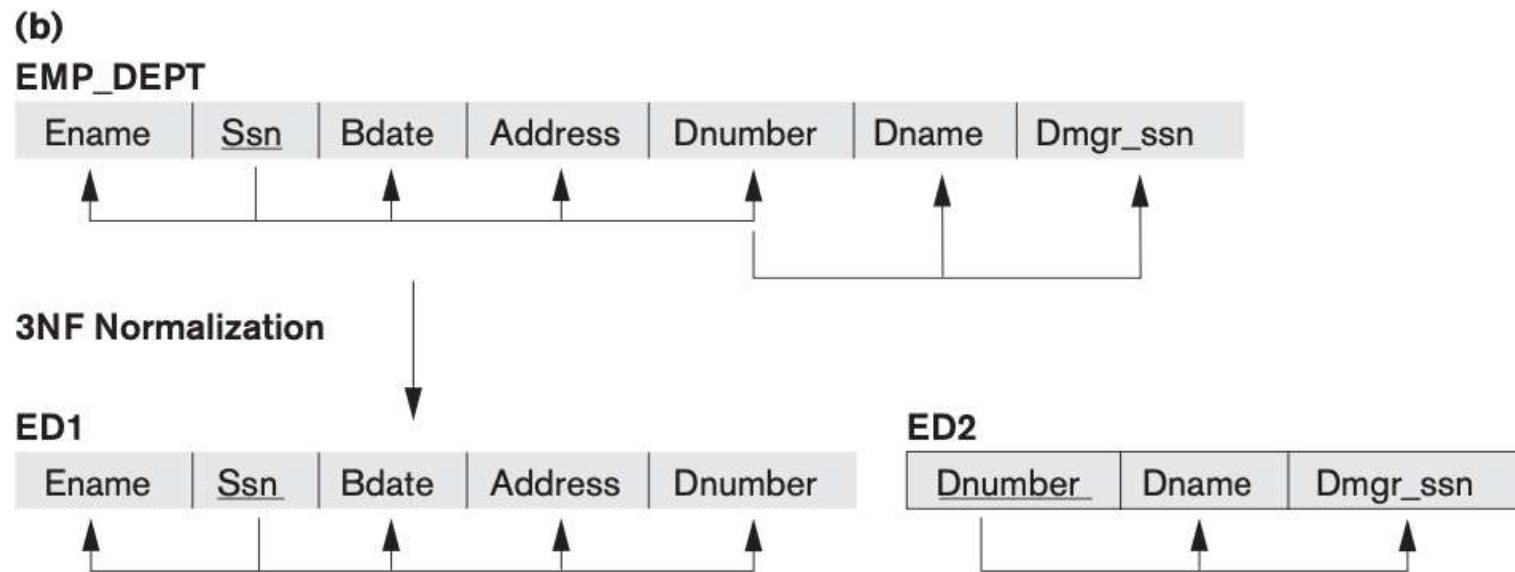
Third Normal Form (3NF)

If we have the following relationship between attributes:

- **A $\rightarrow$ B** (A determines B)
- **B $\rightarrow$ C** (B determines C)
- This means that **A** indirectly determines **C** through **B**, creating a **transitive dependency**
- A table is in **3NF** if and only if both of the following conditions hold:
- The relation R (table) is in second normal form (**2NF**)
- Every non-key attribute of R is **non-transitively dependent** (i.e. directly dependent) on key of R.
- In other words, no non-key attribute should depend on another non-key attribute.

Third Normal Form (3NF)

- Example:



Conditions for a Table to be in 3NF

- A table is in **Third Normal Form (3NF)** if, for every non-trivial functional dependency $X \rightarrow Y$, at least one of the following holds:
- **X is a superkey**: This means that the attribute(s) on the left-hand side of the functional dependency (X) must be a superkey (a key that uniquely identifies a tuple in the table).
- **Y is a prime attribute**: This means that every element of the attribute set Y must be part of a candidate key (i.e., a prime attribute).

Example

- **Relation:** $R(\text{StudentID}, \text{CourseID}, \text{Instructor})$
- **FDs:**
- $(\text{StudentID}, \text{CourseID}) \rightarrow \text{Instructor}$ (a student's course is taught by one instructor)
- $\text{Instructor} \rightarrow \text{CourseID}$ (an instructor teaches only one course)

Test 3NF

- (StudentID, CourseID) $\rightarrow$ Instructor
 - LHS is a superkey $\rightarrow$ OK.
- Instructor $\rightarrow$ CourseID
 - LHS (Instructor) is not a superkey
 - RHS (CourseID) is a prime attribute (part of candidate key)
 - So 3NF allows it.
- Relation is in **3NF**.
- 3NF allows these FD's as conditions are satisfied.
- But
 - Instructor $\rightarrow$ CourseID causes redundancy.
 - We need to write Instructor name for each and every student
 - Also, if Instructor changes course, we'd update multiple rows $\rightarrow$ update anomaly.

Boyce Codd Normal Form (BCNF)

- Sometimes relations in 3NF can still have anomalies and we need a strict rule to normalize the relations.
- **BCNF:** A relation is in BCNF if, for every FD $X \rightarrow Y$:
- X must be a **superkey**.
(no exceptions).

Test BCNF

- In BCNF, every FD must have a **superkey on the left-hand side**.
- Instructor $\rightarrow$ CourseID fails because Instructor is not a superkey.
- So this table is **not in BCNF**.
- **Decompose into BCNF**
- Split into two relations:
- **R1(Instructor, CourseID)** (Instructor determines Course) Now, if an instructor changes from DBMS to AI, we only update **one row** here. No duplication, no anomaly.
- **R2(StudentID, Instructor)** (who teaches each student)
- Now both are in BCNF

Summary

1NF: No repeating groups.

2NF: No partial dependency.

3NF: No transitive dependency